



CHARITH NIROSHAN

**Full-Stack & IoT / AI
Developer**

Giriulla, Sri Lanka | +94 78 864 8969 |
charith123456niroshan@gmail.com

LinkedIn: [www.linkedin.com/in/charith-niroshan-a5a495274]

GitHub:[https://github.com/chrith123456niroshan-rgb]

PROFESSIONAL SUMMARY

Motivated BIT (Hons) student at the University of Vavuniya (GPA: 3.39) with hands-on skills in **Full-Stack Web Development, Computer Vision, and IoT Systems**. Experienced in real-world software testing and agile teamwork through an internship at HIT Innovations. Passionate about using **Python, Java, C#, and IoT hardware** to solve practical problems in an Internship or Junior Software / IoT Engineer role.

TECHNICAL SKILLS

- **Programming Languages:** Java, Python, C#, C++ (Arduino), JavaScript (ES6+), HTML5, CSS3, SQL
- **AI & Computer Vision:** OpenCV, MediaPipe, CVZone
- **Hardware & IoT:** Arduino Microcontrollers, C++ Firmware, Smart Sensors (pH, EC, Temp), Relays
- **Tools & Frameworks:** Git, GitHub, .NET Framework (WinForms), VS Code, Visual Studio
- **Core Competencies:** Systems Analysis & Design (SAD Diagrams), Object-Oriented Programming (OOP), Software Testing & Optimization

SOFT SKILLS

- *Problem Solving & Analytical Thinking*
- *Team Collaboration & Agile Communication*
- *Adaptability & Quick Learning*
- *Project Management & Time Optimization*

LANGUAGES

- ✓ *English:* Professional Working Proficiency
- ✓ *Sinhala:* Native / Full Professional Proficiency

EDUCATION

❖ Bachelor of Information Technology (BIT Hons)

Faculty of Technological Studies, University of Vavuniya | **2023 – Present** (Academic Batch 2022/2023)

- **Current Cumulative GPA: 3.39 / 4.00**
- **Key Coursework:** Web Technologies, Database Management Systems, Software Engineering & SAD, Data Structures & Algorithms, Object-Oriented Programming.

❖ Diploma in Information Communication Technology

IMBS Green Campus | Completed

- **Key Focus:** Computing Fundamentals, Systems Architecture, Applied IT.

❖ G.C.E. Advanced Level (Technology Stream)

Sandalankawa Central College

- **Bio System Technology (BST):** B
- **Science for Technology (SFT):** B
- **Information & Communication Technology (ICT):** C

❖ G.C.E. Ordinary Level

Sandalankawa Central College

- **A Passes (3):** Religion, ICT, Geography
- **B Passes (5):** Mathematics, Science, Sinhala, History, Drama
- **C Pass (1):** English

WORK EXPERIENCE

Technical Intern | HIT Innovations Pvt Ltd *Duration: 2 Months*

- Applied core software engineering, debugging, and testing practices in enterprise software projects.
- Assisted senior engineers in code optimization and agile workflows.
- Conducted system testing and documented software issue trackers.

KEY PROJECTS

Real-Time Cricket Scoring Web App | *HTML5, CSS3, JavaScript, GitHub Pages*

- Built a responsive web app replacing manual scoresheets with real-time ball-by-ball score calculations.
- Integrated dynamic run-rate tracking algorithms and live target indicators.

Gesture-Controlled Snake Game | *Python, OpenCV, MediaPipe, CVZone*

- Developed a computer vision interactive game replacing keyboard inputs with realtime hand gesture tracking.
- Processed multi-point landmark detection for dynamic direction control.

ANSI Color ASCII Video Player | *Python, OpenCV, ANSI Truecolor*

- Built a CLI tool that processes MP4 video streams into 24-bit Truecolor terminal ASCII art in real-time.
- Optimized frame-buffer operations to eliminate terminal rendering flickers.

Smart Hydroponics Grow Tower | *Arduino C++, IoT Sensors, Actuator Relays*

- Designed an automated IoT farming system with real-time pH and EC water quality tracking.
- Programmed closed-loop firmware to dynamically trigger automated nutrient dosing pumps.

Hostel Allocation System | *C#, .NET Framework, Windows Forms*

- Created a desktop administration app automating student room assignments and payment recordkeeping.
- Designed system workflows and relational database architectures using SAD principles.